

The Pickrick Protests: An Interactive Geo AR Experience

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ABSTRACT

With AR's dynamic and immersive qualities, the Pickrick AR project aims to serve two purposes: to support the civil rights movement in remembrance of the African Americans who fought for equality and to educate students about this special moment in history via a pedagogical tool. With considerations of the Covid-19 pandemic, this tool aims to create an immersive and enlightening experience within the real physical space without the presence of a tour guide. In this paper, is an introduction to our design concepts, specifically in how to integrate multimedia information and create a user-friendly interface by using Human Computer Interaction (HCI) principles and the Sense of Place (SOP) Theory.

Index Terms: Human-centered computing—Interaction Design—Interaction design process and methods—User interface design; Social and professional topics—User characteristics—Cultural characteristics

1 INTRODUCTION

The Pickrick AR is a project commemorating the African Americans from the 1964-1965 Pickrick incidents who fought for equal rights. "Pickrick" refers to the name of the restaurant owned by Lester Maddox, former governor of Georgia and staunch segregationist, who notoriously refused service to black people despite the passing of the 1964 Civil Rights Act. Located on Hemphill Avenue, the building was later bought by Georgia Tech (GT) for student services. During the Pickrick incidents, several confrontations occurred between the restaurant owner, surrounding restaurant patrons, and several non-violent black protestors. As the black protestors peacefully attempted to be given service, they were faced with the violent threats and actions of white segregationists using axe handles, guns and baseball bats. Today, the Pickrick has been demolished for the construction of GT's new student center. In order to do these important historical stories justice, our team rejects traditional printed plate commemorations and aims to highlight and educate through a vivid and exciting new form of media. In addition, the recent Covid-19 pandemic calls for a certain safety procedure which makes organized physical tours to the site impractical. In considering several immersive media experiences, we deemed AR most fit to our purposes due to its geolocation feature and its interactive reality enhancement capabilities. Using both these characteristics, visitors of site will be able to "time travel" to the specific dates of the Pickrick incidents and witness what happened with their own eyes. In addendum, contextual information will be provided to supplement the AR experience in order to create a more well-rounded understanding.

Although many studies have been conducted concerning AR technology, there is a distinct lack in studies specifically concerning

AR and user interface. Because of this, our paper will mainly focus on 1) How to make an effective history-topic AR application for educational purposes; 2) How to integrate multiple information into AR and have a user-friendly interface.

2 RELATED WORKS

There are many existing AR applications within museums and/or with historical and educational purpose. In 2017, the Art Gallery of Ontario launched the project ReBlink with digital artist Alex Mayhew. Using an app that scans the surrounding pieces, the AR display showed the characters inside to be alive and moving through waving their hands or blinking their eyes at you. Another example comes from Latvian National Museum who use AR technology to "borrow" world-famous artwork to their museum space. Showcasing several walls printed QR codes which, after scanning, reveals the famous artworks in front of you. While creative, these existing projects have distinct limitations in its fractured, bit-by-bit delivery form. A typical user interaction may consist of scanning a code, receiving information, scanning a different code, receiving the next piece of information, and so on and so forth. As this is inefficient and hinders the flow in creating historical understanding, our project aims to solve this design problem.

3 DESIGN CONCEPTS

3.1 The Sense of Place Theory

The Sense of Place (SOP) is a concept that combines the feeling of attachment, dependence, concern, identity and belongings that people develop regarding a place [1]. There are three sub-parts included in the SOP: place attachment (PA), place dependence (PD), and place identity (PI), "which represent the affective, behavioral, and cognitive relationship, respectively, that people develop with places." [1] In application, SOP serves as an important tool in AR learning and creating personal attachments to a place. Since the Pickrick project includes an AR geolocation application, the SOP theory can guide us in designing for increasing engagement, empathy and retention towards the historical site, the black protestors, and their respective characteristics. According to Yu-Lien Chang et al's [1] AR guidance application and the theory from Tilden (1977) [2] and Beck and Cable (2002) [4], multimedia, such as images, audio, video, text, songs, have proven to be effective [1] in enhancing the SOP. In other words, the SOP theory tells us that providing various types of information related to the place helps visitors increase their sense of place.

Applying the SOP theory to the Pickrick project, we decided to recreate the Pickrick incidents using 3D models and animations in order to best evoke a sense of witnessing the confrontation in person. To supplement this experience will be media such as a historical image of Pickrick, the audio/video displaying the aggressive behavior of the white segregationists, a video of the 1964 Civil Rights Act signing, the video of Attorney Constance Baker Motley in court against the restaurant owner, and the sign when the restaurant finally closed. As these provide significant and relevant historical context concerning the incidents, they are necessary in creating a holistic and richer experience for the user.

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3.2 The general HCI concept for AR application

Keeping in mind the limitation of existing AR applications in its discontinuous and fractured delivery of information, our AR tool aims to provide a continuous and comprehensive historical narrative of the series of civil rights events. More specifically, we will not just simply show the three confrontations by using AR, but also indicate the causal relationships between the confrontations and its relevant historical points (the 1964 Civil Rights Act, the efforts of attorney Constance Baker Motley towards helping the African Americans, the closing of the restaurant, etc.) All these events will be displayed in a connected and chronological manner in order to create a unified experience, rather than disjointed accounts of the most dramatic incidents. Using Dünser's general HCI principles to AR systems design [3] (which includes affordance, minimal cognitive overhead, low physical effort, high learnability, and high flexibility in use), we decided to use the tool "Timeline" in designing our user interface. As a commonly used element in mobile applications, "Timeline" serves several purposes in 1) creating a familiar interface for users in order to increase learnability; 2) integrating and weaving together several pieces of information and multimedia; 3) increasing user freedom in its slide-ability to and from of events.

3.3 Implement Details

Our application is implemented through Win10 with Unity3D (2019.2.21f1) with GPS plugin, AR+GPS Location, made by Daniel Fortes (purchased in the Unity Store). The other packages installed are AR Foundation-preview3-3.0.0, AR Core XR Plugin-preview3-3.0.0, AR Kit XR Plugin-preview3-3.0.0, XR Interaction Toolkit-preview-0.0.6. The mobile is Samsung Galaxy S9 Plus. There are three main parts in our project:

3.3.1 Character Making

In order to accurately recreate the Pickrick incidents to be displayed in AR, research was conducted in collection of various documents, archived videos, images, newspaper and other media. Based off this research, we then modeled the individuals and environment within these events using Blender 2.82. (Figure 1.a)

3.3.2 Timeline Structure

Using the "Timeline" tool to continuously and holistically display the Pickrick incidents and the relevant historical events, we arranged the content chronologically from the restaurant opening (1947) to its closing (1965). The timeline primarily uses a scroll rectangle and buttons to function. The user can then activate the event that is in the center of screen and can drag their finger to modify which event is in the center. When the user releases their finger, the script will find the event closest to the center of the screen, animate this event to the center of the screen, and allow the user click on this event. The user can activate the event on the center of the screen by pressing the picture associated the event (which is a button). If the event you are launching has an "Event-Info" script attached to it, this method will show a page and either an image or video of a certain event. If the event you are launching has an "Event-AR" script attached to it, this method will launch the AR scene associated with this event. (Figure 1.b) (please check our project in Github)¹.

3.3.3 Navigation Sign

Although all Pickrick incidents occurred at the Pickrick restaurant, each incident occurred on slightly different locations on the site. Therefore, a navigation method was needed to guide the interactor to look in the correct direction to see the corresponding models and animations. Thus far we have tried to use a small arrow floating in the AR screen to guide the user towards where they should move. The mechanic is to check whether the vector from the camera to the

¹Github: https://github.com/lingfengzyc/PickrickAR_UI



Figure 1: a. Modeling; b. Timeline and Navigation Design

object is more to the left side or right side. Then if it is on the left side, the arrows would point left and vice versa. (Figure 1.b: the red "Pickrick" icon)

3.3.4 Special Design Consideration for Covid-19

This Pickrick AR app aims to help student keep social distancing guidelines while engaging with the physical site. This is a replacement of traditional tours where there is a physical person as a guide. With this AR App, students would not be forced near one another nor be endangered by the guide's potential infectiousness, instead learning the history independently. Furthermore, in consideration to "stay-at-home" policies, for those who cannot come to the campus, we developed a non-geolocation version and a small-scaled version, enabling users to view the AR content in any empty space (e.g., parking lots, parks, backyards, on the ground at home, etc.). We hope by taking these special design considerations, users of the app can engage with the site as safely as possible.

4 CONCLUSION

As a project developed during the Covid-19 pandemic, we hope the Pickrick AR can be beneficial in providing an experience rival to physical tours. In addition, we aim to support the civil rights movement in showcasing a hidden but important historical story of several Atlanta black individuals who fought for equality. When designing our app, we applied the "sense of place" theory and general HCI principles to integrate a series of historical events. For the user interface, we used a timeline to show this period chronologically with interactive sliders allowing for back-and-forth navigation through time. Inside the app, we applied 3D models and animations to recreate the three most dramatic confrontations. Furthermore, the timeline includes several valuable archived historical videos and images connecting specific dates to our 3D scenes. Through this app, we hope users will be able to connect to history in an engaging and easy manner, all while keeping safe.

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